Prebiotic properties of extruded maize starch-caffeic acid complexes: A study from the small intestine to colon *in vitro*

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Highlights

- Extruded maize starch-caffeic acid complexes had higher level of resistant starch.
- Gut flora fully used the digestive residues of the complexes (REMS-CA).
- REMS-CA produced higher level of total short chain fatty acids.
- REMS-CA promoted the abundance of Bifidobacterium and Lactococcus genus.

Abstract

Recent studies show that starch-polyphenols complexes exert positive effects on gut health, but the probiotic effects of maize starch-caffeic complexes remain underexplored. Therefore, this study aimed to investigate the probiotic effect of maize starch–caffeic complexes from the small intestine to the colon. First, maize starch was extruded with caffeic acid and subjected to *in vitro* digestion, and the undigested parts were fermented *in vitro*, and the structural characteristics, short chain fatty acids (SCFAs) and microbiota communities were investigated. Results showed that caffeic acid reduced the long/short-range order of maize starch after extrusion, significantly increasing resistant starch to 30.35 ± 2.36 %. *In vitro* fermentation indicated that microbiota utilized the amorphous area of the residues first, promoting SCFAs production and the growth of Bifidobacterium and Lactococcus genus. Overall, the probiotic properties of extruded

maize starch–caffeic acid complexes suggest they could serve as a functional food for health benefits.

Introduction

Starch is a very important resource for human beings and provides about 40 %–60 % energy for daily life, since it can be hydrolysed into glucose and absorbed easily (Cai et al., 2024). However, the fast hydrolysis rate of starch into glucose in the human body might induce many risks such as hyperglycaemia, type II diabetes, *etc.* Up to now, many physical, chemical and biological approaches have been used to modify starch to obtain the slower digestion. (Naseer et al., 2021; Wee & Henry, 2020). In particular, there is increasing interest in using phytochemicals to prepare starch-phytochemical complexes to control the digestive process and achieve a perfect staple food (Deng et al., 2021; Fan et al., 2024; Zheng et al., 2020). It has been confirmed that phytochemicals can influence the activity of enzymes as well as the physicochemical properties of starch, and increase the content of slowly digestible starch (SDS) and resistant starch (RS), which can retard the hydrolysis rate of starch *in vitro* (Zheng et al., 2020; Zheng et al., 2021).

Due to the high content of resistant starch in starch-polyphenol complexes, the probiotic properties of resistant starch after digestion in the small intestine warrant attention. Resistant starch can serve as a substrate for beneficial gut bacteria, promoting their growth and activity. In addition, phytochemicals, like flavonoid anthocyanidins and phenolic acids, have been widely investigated as ‘prebiotics’ and modify the ecology of gut microbiota (*Bifidobacterium spp.*, *Lactobacillus spp.* and *Enterococcus spp. etc.*) (Cheng et al., 2023; Lei et al., 2024), and promote the production of short-chain fatty acids (SCFAs), thereby influencing host health (Gowd, Karim, Shishir, Xie, & Chen, 2019). Consequently, starch-polyphenol complexes may not only provide nutritional benefits but also contribute positively to gut microbiota composition and function. More and more studies have shown that starch-polyphenol complexes are also considered as a new type of resistant starch. For example, ferulic acid, caffeic acid, gallic acid, isoquercetin, astragalin, and hyperin were complexed with sweet potato starch and stimulated the growth of *Lactobacillus rhamnosus* and *Bifidobacterium bifidum*, while inhibiting the growth of *Escherichia coli* (Cai et al., 2024). Furthermore, tannic acid can assist with starch to promote the production of SCFAs and the abundance of probiotics (Liu, Luo, Liu, & Hu, 2024). Those findings suggested that the incorporation of polyphenols and starch exert positive effect of gut health.

Given the beneficial effects of RS and phytochemicals in the small intestine, it is essential to investigate how the complexes impact gut health as they transition from the small intestine to the large intestine (Liu, Deng, Luo, Liu, & Hu, 2023; Liu et al., 2023). In our previous study, extrusion, which is green and efficient technology, can enhance the binding of chlorogenic acid to rice starch and increased, increasing the proportion of RS (Zheng et al., 2021). The structure of caffeic acid allows for effective non-covalent interactions with starch, such as hydrogen bonding and π-π stacking, modifying starch's physicochemical properties like solubility and stability. Moreover, caffeic acid is widely available, cost-effective, and adds nutritional and therapeutic benefits to the final product (Cui, Wang, Chen, Mu, & Chen, 2023). Based on this, we hypothesized that extruded maize starch-caffeic acid complexes could increase the level of RS, and then them may be degraded and utilized by intestinal microflora, leading to the production of beneficial substances, such as SCFAs. To further verify this hypothesis, extruded maize starch-caffeic acid complexes were prepared and *in vitro* digestion was conducted. Furthermore, digestive residues of extruded maize starch with or without caffeic acid (REMS-CA and REMS, respectively) were subjected to faecal microbiota. Their structural features, SCFAs level, metabolites of caffeic acid and microbiota composition of the corresponding fermented residues obtained during the *in vitro* fermentation period were determined by SEM, X-ray diffraction, FT-IR, UPLS-TOF-MS^2 and 16S rRNA gene sequencing technology. Our study might provide a theoretical foundation for using extruded maize starch-caffeic acid as functional food substances to improve blood glucose homeostasis and regulate gut health, which may be beneficial to body health.

Materials

Maize starch (moisture content 11.42 %, amylose 22.77 %, amylopectin content 58.91 %) was purchased from Shenyang Leishida Co., Ltd.; caffeic acid (CAS: 331-39-5), methanol (CAS: 67-56-1, HPLC grade), formic acid (CAS: 64-18-6, HPLC grade) and fructo-oligosaccharide (FOS; CAS: 308066-66-2) were bought from Aladdin Co. (Shanghai, China). Pepsin (800–2500 units/mg protein, CAS: 9001-75-6), porcine pancreatic α-amylase (10 U/mg), α-amyloglucosidase (AMG, 10^6 U/mL), invertase (300 U/mg) and

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Maize starch granule broken and gelatinized under extrusion as showed in Fig. 1A, and the addition of caffeic acid reduced the gelatinized degree (Table S2), which may be attributed to the water interaction, and make it difficult for starch to gelatinize. Besides, the binding amount of caffeic acid increased to 12.36 mg/g, it means that about 61.8 % caffeic acid were bond with maize starch, which was lower than that of ferulic acid (Zheng et al., 2024).

Fig. 1B displayed the XRD patterns of the

Conclusions

In the current study, we found that the crystal type of extruded maize starch-caffeic acid complex was A + V type, and caffeic acid reduced the long-range order, while increased the short-range order. Besides, the level of RS was significantly enhanced with the addition of 2.0 %(*w*/w) caffeic acid. Furthermore, REMS and REMS-CA were degraded by large intestinal microbiota and influenced the microflora communities. The catabolism was different in Caffeic acid and REMS-CA group due to the

CRediT authorship contribution statement

Yuxue Zheng: Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Xingqian Ye:** Resources, Project administration, Funding acquisition. **Yanyu Hu:** Writing – review & editing, Software, Investigation. **Shaoyun Wang:** Writing – review & editing, Supervision, Project administration. **Jinhu Tian:** Resources, Project administration, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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